LEARNING MODULE

28

Mean–Variance Optimization

Minimize variance for a target mean in closed form, find the best Sharpe ratio beside a risk-free asset, and measure how far the answers move with their inputs.

VOLUME V · Portfolio Optimization

BUILDS ON LM7 · LM8 · LM9 · LM10 · LM11 · LM12 · LM16 · LM20 · LM23 · LM27

LEADS TO LM29 · LM30 · LM31 · LM32 · LM33 · LM34 · LM35 · LM43

TIME ≈ 2.5 h + lab

LEARNING OUTCOMES

Mastery	*After this module you should be able to:*
☐	**1** derive the minimum-variance portfolio of N assets from the Lagrangian, and state when it exists and is unique
☐	**2** derive the frontier as a parabola in variance and mean, and explain two-fund separation
☐	**3** derive the tangency portfolio and the capital market line, and state when the tangency portfolio exists
☐	**4** compute a risk-averse investor's exposures beside a risk-free asset, and relate them to the tangency portfolio
☐	**5** compute how tangency weights respond to one mean, and compare that change with the mean's standard error
☐	**6** split the gap between predicted and realized volatility into a moved minimum and a weight error
☐	**7** compute frontier portfolios in NumPy from linear systems, and check them and their multipliers with cvxpy
☐	**8** explain why unconstrained mean–variance weights are in-sample estimates, not recommendations

LEARNING MODULE OVERVIEW

- For a positive definite Σ, minimum variance under linear constraints has one closed-form solution, (28.1). On E5's first window the minimum-variance mix holds all five funds and predicts 5.02% a year.
- The frontier is a parabola in variance and mean. Its asymptotes have slope $\mathrm{SR}_\infty = 0.675$, and any two frontier portfolios rebuild it.
- The tangency portfolio $\Sigma^{-1}\mathbf{e}/(\mathbf{1}^\top\Sigma^{-1}\mathbf{e})$ exists only when $r_f < \mu_{\mathrm{mv}}$. On the first window its Sharpe ratio is 0.880, with oil sold short. In 22 of the through case's 92 windows it does not exist.
- A risk-averse investor holds $\Sigma^{-1}\mathbf{e}/\delta$, the tangency mix levered or diluted with cash: a 10% volatility needs $\delta = 8.80$ and exposures of 1.28 times capital.
- One more point of expected return on gold, 0.15 of its standard error, moves gold's tangency weight from 2.6% to 12.8%.
- Held to October 2026, the two portfolios realized 10.05% and 14.15% volatility against 5.02% and 7.81% predicted. The test period's own minimum on its joint dates was 9.63%, and the weight errors added 0.41 and 4.51 points to it (10.04% and 14.14% there).
- On 28 sector indices the minimum-variance portfolio shorts 12 of them, with gross exposure 2.24.

1 INTRODUCTION

LM27 took the weights $\mathbf{w}$ as given and computed a portfolio's mean, variance and risk contributions. This module chooses $\mathbf{w}$: the least variance for a target mean, and the best Sharpe ratio beside a risk-free asset. It also states when each answer exists and how far it moves with its inputs. Choosing weights by mean and variance began with Markowitz (1952).

MOTIVATING CASE **Five funds, two years, one optimizer**

At the end of February 2019 a TWD investor holds five Taiwan-listed funds. They track Taiwan stocks (0050), US stocks (00646), long US Treasuries (00679B), gold futures (00635U) and crude-oil futures (00642U). The investor has 504 days of joint returns, from 13 February 2017 to 27 February 2019. Which mix has the least variance, and which the best Sharpe ratio?

On those days the calculus is exact. The minimum-variance mix holds every fund, 41.4% in the bond fund, and predicts 5.02% volatility a year. At a 1.5% risk-free rate the best Sharpe ratio is 0.88, at 7.81% volatility, with oil sold short (Exhibit 1).

Hold both mixes from 4 March 2019 to 8 October 2026. The first realizes 10.05%, twice its prediction; the second 14.15%. And one more point of expected return on gold, under a sixth of its standard error, would lift gold's weight from 2.6% to 12.8%. Was the calculus wrong, were the weights wrong, or did the inputs change?

Exhibit 1: E5's first window, its two optimal portfolios and the test period

Fund	Window mean (%)	Window volatility (%)	Test volatility (%)	$\mathbf{w}^{\mathrm{mv}}$ (%)	$\mathbf{w}^{\tan}$ (%)
0050	7.5	13.5	21.6	10.7	11.8
00646	10.0	11.9	17.2	22.5	62.6
00679B	2.6	8.5	14.6	41.4	37.8
00635U	0.8	9.4	18.1	25.0	2.6
00642U	−1.4	23.2	40.7	0.4	−14.7

Portfolio	Predicted volatility (%)	Realized volatility (%)	Predicted mean (%)	Realized mean (%)
$\mathbf{w}^{\mathrm{mv}}$	5.02	10.05	4.34	9.14
$\mathbf{w}^{\tan}$	7.81	14.15	8.38	10.79

Note: daily total returns in NT$, a year; $r_f = 1.5\%$. The window's columns are the optimizer's inputs on its 504 joint dates, annualized by $A = 242.95$. The test period runs from 5 March 2019 to 8 October 2026.

Each test volatility uses the fund's own one-session dates and its own sessions per year: $A_{\mathrm{w}} = 243.57$, or $A = 242.95$ for 0050. On a common A they differ from the diagonal of the 1,845 test rows' joint matrix by at most 0.07 points (00642U). The realized columns are out-of-sample, on the wealth frame at constant weights, without costs. The weights are in-sample plug-in estimates, and the $\mathbf{w}^{\tan}$ column, rounded separately, sums to 100.1.

Source: FinMind `TaiwanStockPrice`, `TaiwanStockDividendResult` and `TaiwanStockSplitPrice`, retrieved 24 October 2026; `data/vol5_series.py`. Computations: `data/lm28_numbers.py`.

The first window was calm. Its minimum volatility ranks third lowest of the through case's 92 windows, whose minima run from 4.97% (31 May 2019) to 10.42% (31 March 2022).

Sections 2 to 4 derive the minimum-variance portfolio, the frontier and the tangency portfolio. Section 5 measures how the weights react to the means, Section 6 holds the first window's portfolios to the end, and Section 7 computes in Python.

NOTE **Notation for this module**

E5 is LM27's five funds, in the order 0050, 00646, 00679B, 00635U, 00642U, with daily total returns in NT$. Its estimation frame holds 2,362 joint returns from 18 January 2017 to 8 October 2026, with $A = 242.95$; its wealth frame holds all 2,368 sessions, with $A_{\mathrm{w}} = 243.57$. The examples set $\boldsymbol{\mu} := A\bar{\mathbf{R}}$ and $\Sigma := A\hat{\Sigma}$ (divisor T) on a window and treat these plug-in estimates as the model's inputs; the theory keeps $\boldsymbol{\mu}$ and Σ unhatted, as LM7 did. The risk-free rate is an illustrative $r_f = 1.5\%$ a year, simple, or 0.617 bp a day, and $\mathbf{e} = \boldsymbol{\mu} - r_f\mathbf{1}$ with entries e_i. Weights print in percent, a short is a weight below -10^{-6} as a fraction of capital, and gross exposure is $\|\mathbf{w}\|_1$, as a multiple of capital.

Local: μ_{mv}, $\mu_{\tan}$, $\sigma_{\tan}$ the minimum-variance portfolio's mean and the tangency portfolio's mean and volatility; SR_∞ the slope of the frontier's asymptotes (new: scalars a, b, c, d for the frontier would clash with a bet's loss and gain, the Kelly fraction and LM23's number of coefficients); $\mathbf{w}_a$, $\mathbf{w}_b$ and μ_a, μ_b two distinct frontier portfolios and their means (Section 3); K ($k \times N$) and $\mathbf{b}$ the constraints $K\mathbf{w} = \mathbf{b}$, k their number; $\boldsymbol{\lambda}$ their multipliers, with entries λ (budget) and λ_μ (target mean), so that λ_1 and λ_N stay eigenvalues; $\boldsymbol{\Delta}$ a weight change that keeps the constraints, $K\boldsymbol{\Delta} = \mathbf{0}$; $\Delta\mathbf{e}$ a change of the excess returns; $\delta > 0$ risk aversion; $\mathbf{f}^*$ the optimal exposures of the problem being solved, Section 4's least-variance book or the maximum of (28.8) (mean–variance optima; the Kelly fractions of N assets are LM43's); $L = \mathbf{1}^\top\mathbf{f}$ the total exposure; $\beta_{i,p}$ LM27's beta of fund i to portfolio p. Counts: T returns (504, 1,845, 1,851, 4,382), N assets (5 or 28), i and j funds.

NOTE **Research hygiene**

Sections 2 to 5 treat the first window's sample moments as known: every weight there is **in-sample**. Section 6 holds two portfolios, chosen before any result, from 4 March 2019 to 8 October 2026 without costs. That single test is **out-of-sample**. The test period's own frontier is in-sample on later data and only explains the miss.

The scan of the 92 windows is **descriptive**. The risk aversions, the 10% volatility target, the 1-point changes of a mean and the rates 0% and 2% are **hypothetical**. U28's figures are price indices, covariance only. No portfolio here is a rule of the through case or a candidate for its best rule (LM33). Weights and frontiers are plug-in estimates with an unmeasured error. Section 5 shows how far the weights move with one mean, and LM30 measures the error.

2 THE MINIMUM-VARIANCE PORTFOLIO

- [ ] derive the minimum-variance portfolio of N assets from the Lagrangian, and state when it exists and is unique

LM10 showed that a portfolio's variance $\mathbf{w}^\top\Sigma\mathbf{w}$ is a convex function of the weights, and stopped at its gradient. Minimizing it under linear constraints takes one Lagrangian, as in LM7, now with N assets and k constraints at once.

Let Σ be an $N \times N$ positive definite matrix, K a $k \times N$ matrix of full row rank k, and $\mathbf{b}$ a vector of length k. The feasible set $\{\mathbf{w} : K\mathbf{w} = \mathbf{b}\}$ is not empty, because K has independent rows. The budget alone is $K = \mathbf{1}^\top$ with $\mathbf{b} = 1$.

With LM7's signs the Lagrangian is $\mathcal{L} = \mathbf{w}^\top\Sigma\mathbf{w} - \boldsymbol{\lambda}^\top(K\mathbf{w} - \mathbf{b})$. By (10.8) its gradient in $\mathbf{w}$ is $2\Sigma\mathbf{w} - K^\top\boldsymbol{\lambda}$, so a critical point has $\mathbf{w} = \frac{1}{2}\Sigma^{-1}K^\top\boldsymbol{\lambda}$, and the constraint then requires $\frac{1}{2}K\Sigma^{-1}K^\top\boldsymbol{\lambda} = \mathbf{b}$. The matrix $K\Sigma^{-1}K^\top$ is positive definite. By (11.8) the eigenvalues of Σ^{-1} are the reciprocals of Σ's, all positive, and $K^\top\boldsymbol{\lambda} \neq \mathbf{0}$ whenever $\boldsymbol{\lambda} \neq \mathbf{0}$, because $K^\top$ has independent columns. Hence every $\boldsymbol{\lambda} \neq \mathbf{0}$ has

$$\boldsymbol{\lambda}^\top K\Sigma^{-1}K^\top\boldsymbol{\lambda} = (K^\top\boldsymbol{\lambda})^\top\Sigma^{-1}(K^\top\boldsymbol{\lambda}) > 0,$$

so $K\Sigma^{-1}K^\top\boldsymbol{\lambda} \neq \mathbf{0}$. The matrix is therefore invertible, and the multipliers are unique.

THEOREM **Minimum variance under linear constraints**

Let Σ be positive definite and K ($k \times N$) of full row rank. Then the problem $\min \mathbf{w}^\top\Sigma\mathbf{w}$ subject to $K\mathbf{w} = \mathbf{b}$ has the unique solution $\mathbf{w}^*$, with value and multipliers

$$\begin{aligned} \mathbf{w}^* &= \Sigma^{-1}K^\top\big(K\Sigma^{-1}K^\top\big)^{-1}\mathbf{b}, \\ \mathbf{w}^{*\top}\Sigma\mathbf{w}^* &= \mathbf{b}^\top\big(K\Sigma^{-1}K^\top\big)^{-1}\mathbf{b}, \\ \boldsymbol{\lambda} &= 2\big(K\Sigma^{-1}K^\top\big)^{-1}\mathbf{b}. \end{aligned} \tag{28.1}$$

Every feasible $\mathbf{w}$ satisfies

$$\begin{aligned} \mathbf{w}^\top\Sigma\mathbf{w} = \mathbf{w}^{*\top}\Sigma\mathbf{w}^* \\ + (\mathbf{w} - \mathbf{w}^*)^\top\Sigma(\mathbf{w} - \mathbf{w}^*). \end{aligned}$$

Proof. Let $\mathbf{w}$ be feasible and $\boldsymbol{\Delta} := \mathbf{w} - \mathbf{w}^*$, so that $K\boldsymbol{\Delta} = \mathbf{0}$. By (10.9),

$$\mathbf{w}^\top\Sigma\mathbf{w} = \mathbf{w}^{*\top}\Sigma\mathbf{w}^* + 2\boldsymbol{\Delta}^\top\Sigma\mathbf{w}^* + \boldsymbol{\Delta}^\top\Sigma\boldsymbol{\Delta}.$$

The middle term is $\boldsymbol{\Delta}^\top K^\top\boldsymbol{\lambda} = (K\boldsymbol{\Delta})^\top\boldsymbol{\lambda} = 0$, because $2\Sigma\mathbf{w}^* = K^\top\boldsymbol{\lambda}$. The last term is positive unless $\boldsymbol{\Delta} = \mathbf{0}$, since Σ is positive definite. So $\mathbf{w}^*$ is the unique minimum, and substituting it gives the value. ■

The identity proves existence and uniqueness by itself, so no KKT theory is needed. The result is also LM10's theorem at work: a positive definite Σ makes the variance strictly convex. Long-only weights add inequality constraints, which need the KKT conditions of LM7's (7.13). LM29 applies them.

With the budget alone, $K\Sigma^{-1}K^\top = \mathbf{1}^\top\Sigma^{-1}\mathbf{1}$ is a number, and (28.1) becomes the following result.

RESULT **The global minimum-variance portfolio**

For Σ positive definite, the fully invested portfolio of least variance is

$$\begin{aligned} \mathbf{w}^{\mathrm{mv}} &= \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^\top\Sigma^{-1}\mathbf{1}}, \qquad \sigma^2_{\min} = \frac{1}{\mathbf{1}^\top\Sigma^{-1}\mathbf{1}}, \\ \lambda^* &= 2\sigma^2_{\min}, \qquad \Sigma\mathbf{w}^{\mathrm{mv}} = \sigma^2_{\min}\mathbf{1}. \end{aligned} \tag{28.2}$$

For every $\mathbf{w}$ with $\mathbf{1}^\top\mathbf{w} = 1$,

$$\begin{aligned} \mathbf{w}^\top\Sigma\mathbf{w} = \sigma^2_{\min} \\ + (\mathbf{w} - \mathbf{w}^{\mathrm{mv}})^\top\Sigma(\mathbf{w} - \mathbf{w}^{\mathrm{mv}}). \end{aligned} \tag{28.3}$$

Identity (28.3) is LM7's (7.9) for N assets, and λ^* is its shadow price of the budget. The last part of (28.2) says that every fund has covariance $\sigma^2_{\min}$ with $\mathbf{w}^{\mathrm{mv}}$. So does every fully invested portfolio, since $\mathbf{w}^\top\Sigma\mathbf{w}^{\mathrm{mv}} = \sigma^2_{\min}\mathbf{1}^\top\mathbf{w}$. By (27.5) every fund's beta to $\mathbf{w}^{\mathrm{mv}}$ is 1, and by (27.6) every risk share equals its weight. LM11 solved $\hat{\Sigma}\mathbf{x} = \mathbf{1}$; rescaled to sum to 1, its solution is (28.2).

EXAMPLE 1

The minimum-variance mix of the first window

On E5's first window, solve $\Sigma\mathbf{x} = \mathbf{1}$, rescale the solution, and check its covariances with the funds.

SOLUTION

The rescaled solution is $\mathbf{w}^{\mathrm{mv}} = (10.7, 22.5, 41.4, 25.0, 0.4)\%$. The sum $\mathbf{1}^\top\mathbf{x}$ gives $\sigma^2_{\min} = 0.0025152$, so $\sigma_{\min} = 5.02\%$; also $\mu_{\mathrm{mv}} = 4.34\%$ and $\lambda^* = 0.00503$. The variance carries seven decimals because $\sqrt{0.002515}$ would give 5.01%.

Every fund's covariance with the mix is 0.0025152, so each beta to it is 1 and each risk share equals its weight. The 5.02% lies below every fund's volatility, the lowest being 00679B's 8.5%.

The smallest weight is oil's, 0.36%. In none of the through case's 92 windows does an unconstrained weight turn negative; the smallest is 0.13%. So the long-only bounds of R3, the through case's minimum-variance rule, never bind on E5 (LM29).

EXAMPLE 2

Minimum variance on 28 sector indices

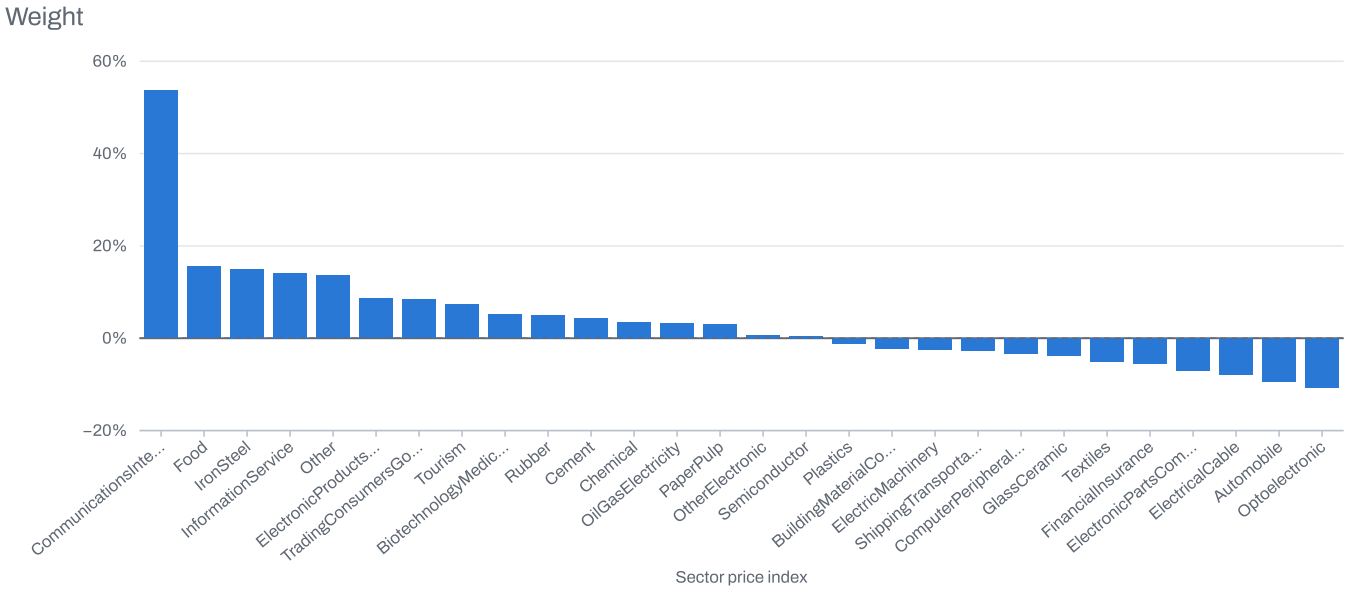
Apply (28.2) to U28, the 28 sector price indices: 4,382 returns from 2 December 2008 to 8 October 2026, $A = 245.48$.

SOLUTION

The portfolio shorts 12 sectors, with gross exposure 2.24 (Exhibit 2). Its volatility is 10.98%, against 15.75% for equal weights (LM10) and 13.67% for the least volatile sector, CommunicationsInternet, which takes 53.6% of the weight. The largest short is Optoelectronic's, −10.8%, and the condition number of Σ is $\kappa = 108.1$.

On LM11's halves of 2,191 returns the portfolio shorts 10 and 13 sectors, with gross exposures 2.44 and 2.06. The cosine of the two weight vectors is LM11's 0.690, because rescaling leaves a cosine unchanged. Both are in-sample. How such weights fare on the other half is LM30's question.

Exhibit 2: Minimum-variance weights of the 28 sector indices



Note: (28.2) on the full sample's $\Sigma = A\hat{\Sigma}$, $A = 245.48$, sorted. Twelve weights are negative, the gross exposure is 2.24 and $\sigma_{\min} = 10.98\%$. Price indices, no dividends; in-sample; covariance only. Source: FinMind `TaiwanStockPrice` (the 28 sector price indices, retrieved 11 October 2026, with LM10's repairs from `TaiwanStockEvery5SecondsIndex`). Computations: `data/lm28_charts.py`.

PITFALL **Solve, never invert, and test positive definiteness first**

Compute $\Sigma^{-1}\mathbf{1}$ with `np.linalg.solve(S, ones)`, which factorizes once and rounds less than forming the inverse. Before that, `np.linalg.cholesky(S)` tests positive definiteness: a symmetric matrix is positive definite exactly when it has a Cholesky factor (LM12). If the factor fails, some nonzero weight vector has zero or negative variance under Σ. Then (28.1) no longer applies, and a unique minimum need not exist.

導讀 **最小變異數：一個線性方程組**

在權重總和為 1 的條件下讓變異數（variance）最小，用的就是 LM7 的拉格朗日乘數（Lagrange multiplier）法，只是資產從兩檔變成 N 檔。共變異數矩陣（covariance matrix）Σ 正定（positive definite）時，變異數是嚴格凸函數（strictly convex function），所以最小值至多只有一個，這就是 LM10 所說最佳化的基礎。最小變異數投資組合（minimum-variance portfolio）是把 $\Sigma\mathbf{x} = \mathbf{1}$ 的解縮放成總和為 1，每一檔基金與它的共變異數都等於 $\sigma^2_{\min}$。恆等式（identity）(28.3) 直接證明它就是全域最小值（global minimum）：最小值存在，而且只有這一個，不需要 KKT 理論。E5 第一個視窗（window）的最小變異數投資組合五檔都持有，預測波動率（volatility）5.02%，低於任何一檔基金。

KNOWLEDGE CHECK 1

Let λ_1 and λ_N be the largest and smallest eigenvalues of Σ. Show that $\lambda_N/N \le \sigma^2_{\min} \le \lambda_1/N$, using (11.4) and $\|\mathbf{w}\|^2 \ge 1/N$ for $\mathbf{1}^\top\mathbf{w} = 1$.

ANSWER

For $\mathbf{w} \neq \mathbf{0}$, (11.4) with $\mathbf{u} = \mathbf{w}/\|\mathbf{w}\|$ gives $\lambda_N\|\mathbf{w}\|^2 \le \mathbf{w}^\top\Sigma\mathbf{w} \le \lambda_1\|\mathbf{w}\|^2$. By the Cauchy–Schwarz inequality (LM8), $1 = (\mathbf{1}^\top\mathbf{w})^2 \le N\|\mathbf{w}\|^2$, so every fully invested $\mathbf{w}$ has variance at least λ_N/N. Equal weights have $\|\mathbf{w}\|^2 = 1/N$, so their variance is at most λ_1/N, and $\sigma^2_{\min}$ is no larger. On the first window, in volatilities: $3.18\% \le 5.02\% \le 7.38\% \le 10.69\%$, the third being equal weights.

3 THE FRONTIER AND TWO-FUND SEPARATION

- ☐ derive the frontier as a parabola in variance and mean, and explain two-fund separation

Now add a target mean. A frontier portfolio has the least variance among fully invested portfolios with mean μ_p: (28.1) with $K = (\mathbf{1}\ \boldsymbol{\mu})^\top$ and $\mathbf{b} = (1, \mu_p)^\top$. The two rows are independent exactly when $\boldsymbol{\mu}$ is not proportional to $\mathbf{1}$.

A shorter route starts from $\mathbf{w}^{\text{mv}}$. Write $\mathbf{w} = \mathbf{w}^{\text{mv}} + \boldsymbol{\Delta}$ with $\mathbf{1}^\top\boldsymbol{\Delta} = 0$. By (28.3) the variance is $\sigma^2_{\min} + \boldsymbol{\Delta}^\top\Sigma\boldsymbol{\Delta}$, and the mean is $\mu_{\text{mv}} + (\boldsymbol{\mu} - \mu_{\text{mv}}\mathbf{1})^\top\boldsymbol{\Delta}$. So $\boldsymbol{\Delta}$ solves (28.1) with the rows $\mathbf{1}^\top$ and $(\boldsymbol{\mu} - \mu_{\text{mv}}\mathbf{1})^\top$ and the right-hand side $(0, \mu_p - \mu_{\text{mv}})^\top$. Its 2×2 matrix is diagonal, because (28.2) gives $\mathbf{1}^\top\Sigma^{-1}\boldsymbol{\mu} = \mu_{\text{mv}}/\sigma^2_{\min}$ and $\mathbf{1}^\top\Sigma^{-1}\mathbf{1} = 1/\sigma^2_{\min}$:

$$\mathbf{1}^\top\Sigma^{-1}(\boldsymbol{\mu} - \mu_{\text{mv}}\mathbf{1}) = \frac{\mu_{\text{mv}}}{\sigma^2_{\min}} - \frac{\mu_{\text{mv}}}{\sigma^2_{\min}} = 0.$$

The inverse of that diagonal matrix follows from (9.5) at once, and (28.1) gives the frontier.

RESULT **The minimum-variance frontier**

Let Σ be positive definite and $\boldsymbol{\mu}$ not proportional to $\mathbf{1}$, and let $\mathrm{SR}_\infty^2 := (\boldsymbol{\mu} - \mu_{\mathrm{mv}}\mathbf{1})^\top \Sigma^{-1}(\boldsymbol{\mu} - \mu_{\mathrm{mv}}\mathbf{1}) > 0$. The fully invested portfolio of least variance with mean μ_p is

$$\mathbf{w}(\mu_p) = \mathbf{w}^{\mathrm{mv}} + \frac{\mu_p - \mu_{\mathrm{mv}}}{\mathrm{SR}_\infty^2}\,\Sigma^{-1}(\boldsymbol{\mu} - \mu_{\mathrm{mv}}\mathbf{1}), \tag{28.4}$$

and its variance is

$$\sigma_p^2 = \sigma_{\min}^2 + \frac{(\mu_p - \mu_{\mathrm{mv}})^2}{\mathrm{SR}_\infty^2}. \tag{28.5}$$

Over all targets, the frontier portfolios form the **minimum-variance frontier**. Equation (28.5) is a parabola in (σ_p^2, μ_p) and a hyperbola in (σ_p, μ_p) with vertex $(\sigma_{\min}, \mu_{\mathrm{mv}})$. As σ_p grows, it approaches the asymptotes $\mu_p = \mu_{\mathrm{mv}} \pm \mathrm{SR}_\infty \sigma_p$ (Problem 8). The upper half, $\mu_p \geq \mu_{\mathrm{mv}}$, is the **efficient frontier**: for each variance it holds the higher mean.

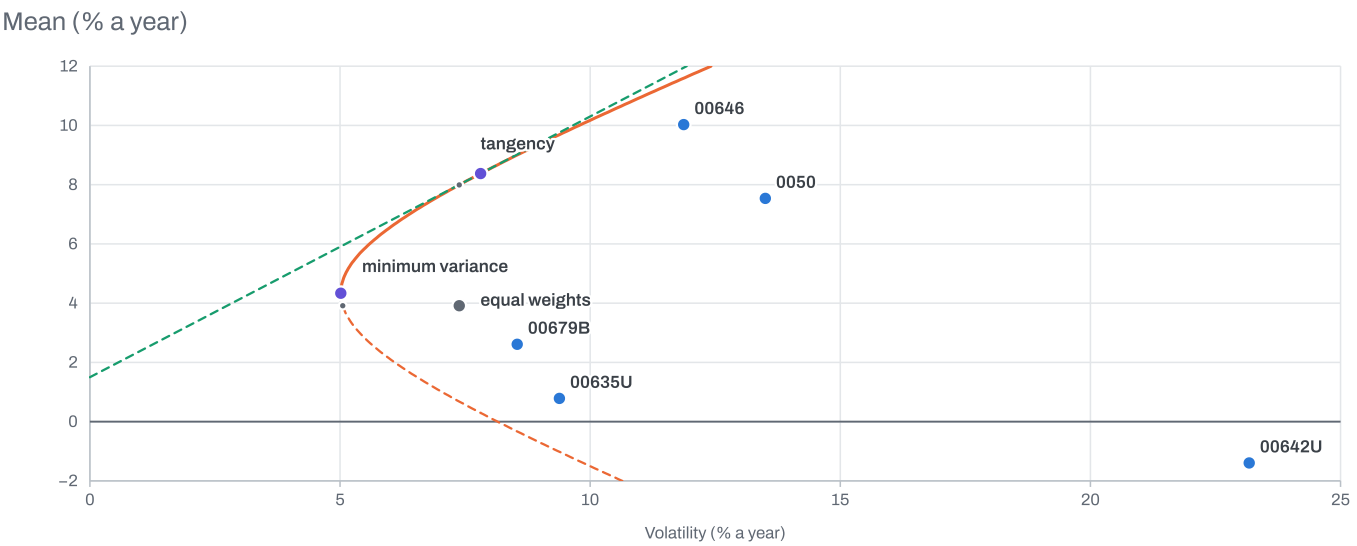
The multiplier of the target mean is a shadow price, (7.6): by (28.5), $\lambda_\mu = d\sigma_p^2/d\mu_p = 2(\mu_p - \mu_{\mathrm{mv}})/\mathrm{SR}_\infty^2$. Deep Dive 1 computes both multipliers from the 2×2 system.

By (28.4) the frontier weights are affine in μ_p. So any two distinct frontier portfolios $\mathbf{w}_a$ and $\mathbf{w}_b$, with means μ_a and μ_b, rebuild every other:

$$\mathbf{w}(\mu_p) = \mathbf{w}_a + \frac{\mu_p - \mu_a}{\mu_b - \mu_a}\,(\mathbf{w}_b - \mathbf{w}_a).$$

The rebuilding is **two-fund separation**: an investor who wants a frontier portfolio needs only two funds. Merton (1972) derived the frontier in closed form and proved this separation as a mutual-fund theorem.

Exhibit 3: The first window's frontier, tangency portfolio and capital market line



Note: in-sample, the first window's plug-in moments, % a year. Solid: the efficient frontier, the upper half of the minimum-variance frontier (28.5). Dashed: its lower half, and the capital market line (28.7) from $r_f = 1.5\%$ through the tangency portfolio, which assumes borrowing at r_f. The asymptotes' slope is $\mathrm{SR}_\infty = 0.675$. Two small points mark the frontier portfolios with equal weights' mean (volatility 5.05%) and with their volatility (mean 7.99%). Source: as Exhibit 1. Computations: `data/lm28_charts.py`.

EXAMPLE 3

Equal weights lie inside the frontier

Place equal weights against the first window's frontier (Exhibit 3).

SOLUTION

Equal weights have mean 3.913% and volatility 7.38%. Their mean lies 0.42 points below $\mu_{\mathrm{mv}} = 4.335\%$, so $\mathbf{w}^{\mathrm{mv}}$ has both the higher mean and the lower volatility. With $\mathrm{SR}_\infty^2 = 0.4554$, (28.5) gives the frontier portfolio with equal weights' mean a volatility of

$$\sqrt{0.0025152 + 0.0042^2/0.4554} = 5.05\%.$$

The efficient portfolio with equal weights' volatility has mean 7.99% and weights $(11.7, 58.8, 38.1, 4.7, -13.3)\%$.

At that efficient point the multipliers are $\lambda = -0.00193$ and

$$\lambda_\mu = \frac{d\sigma_p^2}{d\mu_p} = \frac{2 \times 3.6565\%}{0.4554} = 0.161.$$

By (28.4), each extra point of target mean moves 00646's weight by +9.9 points and gold's by −5.6.

PITFALL **Nearly equal means make a flat frontier**

In (28.4) the weights move by $(\mu_p - \mu_{\text{mv}})/\text{SR}_\infty^2$ times a fixed direction. When the means are nearly equal, SR_∞ is small, and a modest target mean demands huge long and short positions.

DEEP DIVE **Deep Dive 1: the frontier's multipliers from the 2×2 system** optional · click to expand

With $K = (\mathbf{1}\ \boldsymbol{\mu})^\top$, expanding the definition of SR_∞^2 gives $\boldsymbol{\mu}^\top \Sigma^{-1} \boldsymbol{\mu} = \text{SR}_\infty^2 + \mu_{\text{mv}}^2/\sigma_{\min}^2$, so the matrix of (28.1) is

$$K\Sigma^{-1}K^\top = \frac{1}{\sigma_{\min}^2}\begin{pmatrix} 1 & \mu_{\text{mv}} \\ \mu_{\text{mv}} & \mu_{\text{mv}}^2 + \sigma_{\min}^2 \text{SR}_\infty^2 \end{pmatrix}.$$

Its determinant is $\text{SR}_\infty^2/\sigma_{\min}^2$, positive exactly when $\boldsymbol{\mu}$ is not proportional to $\mathbf{1}$. Cramer's rule (9.4) solves $\frac{1}{2}K\Sigma^{-1}K^\top \boldsymbol{\lambda} = (1, \mu_p)^\top$:

$$\lambda_\mu = \frac{2(\mu_p - \mu_{\text{mv}})}{\text{SR}_\infty^2}, \qquad \lambda = 2\sigma_{\min}^2 - \mu_{\text{mv}}\lambda_\mu.$$

Then $\mathbf{w} = \frac{1}{2}\Sigma^{-1}(\lambda \mathbf{1} + \lambda_\mu \boldsymbol{\mu})$ is (28.4), and the value $\frac{1}{2}(\lambda + \lambda_\mu \mu_p)$ is (28.5).

導讀 **最小變異數前緣是一條拋物線**

指定目標平均報酬（mean return）之後，變異數（variance）最小的投資組合構成最小變異數前緣（minimum-variance frontier）。以變異數為橫軸、平均數（mean）為縱軸時它是一條拋物線（parabola），換成波動率（volatility）為橫軸則是雙曲線（hyperbola），頂點是最小變異數投資組合（minimum-variance portfolio），兩條漸近線（asymptote）的斜率是 $\pm\text{SR}_\infty$。前緣上的權重是目標平均數的一次函數，所以任選兩個前緣投資組合，就能組合出其他所有的前緣投資組合：這就是兩基金分離（two-fund separation），是分離定理（separation theorem）的一種。上半支才是效率前緣（efficient frontier），上面的投資組合是效率投資組合（efficient portfolio）。第一個視窗（window）的等權重投資組合落在前緣內側，平均數比最小變異數投資組合低，波動卻更高。

KNOWLEDGE CHECK 2

Why is there no frontier when every fund has the same mean? And why is equal weighting inefficient on the first window?

ANSWER

Then every fully invested portfolio has that same mean, and $K = (\mathbf{1}\ \boldsymbol{\mu})^\top$ has rank 1. No other target is feasible, $\mathrm{SR}_\infty = 0$, and (28.5) divides by zero. Equal weights' mean, 3.913%, lies 0.42 points below $\mu_{\mathrm{mv}} = 4.335\%$, at a higher volatility.

4 THE TANGENCY PORTFOLIO AND THE CAPITAL MARKET LINE

- ☐ derive the tangency portfolio and the capital market line, and state when the tangency portfolio exists
- ☐ compute a risk-averse investor's exposures beside a risk-free asset, and relate them to the tangency portfolio

Now add a risk-free asset that earns r_f. A book holds exposures $\mathbf{f}$ to the funds and cash $1 - L$ at r_f, where $L = \mathbf{1}^\top \mathbf{f}$, so cash is borrowing when $L > 1$. By (8.4) its return is $r_f + \mathbf{f}^\top(\mathbf{R} - r_f\mathbf{1})$, with mean $r_f + \mathbf{f}^\top\mathbf{e}$ and, by (16.4), variance $\mathbf{f}^\top\Sigma\mathbf{f}$. Borrowing without limit at the risk-free rate is an idealization. LM29 adds a financing spread and limits.

The least-variance book with mean $\mu_p \neq r_f$ solves (28.1) with $K = \mathbf{e}^\top$ and $\mathbf{b} = \mu_p - r_f$, for any $\mathbf{e} \neq \mathbf{0}$:

$$\mathbf{f}^* = \frac{(\mu_p - r_f)\,\Sigma^{-1}\mathbf{e}}{\mathbf{e}^\top\Sigma^{-1}\mathbf{e}}, \qquad \mathrm{SR}^{*2} := \mathbf{e}^\top\Sigma^{-1}\mathbf{e},$$
$$\mathbf{f}^{*\top}\Sigma\mathbf{f}^* = \frac{(\mu_p - r_f)^2}{\mathrm{SR}^{*2}}.$$

So every least-variance book is a multiple of $\Sigma^{-1}\mathbf{e}$, and every book has $|\mu_p - r_f| \le \mathrm{SR}^*\sigma_p$. The least-variance books with $\mu_p > r_f$, the efficient ones, lie on the **capital market line** $\mu_p = r_f + \mathrm{SR}^*\sigma_p$.

A fully invested book on that direction divides $\Sigma^{-1}\mathbf{e}$ by $\mathbf{1}^\top\Sigma^{-1}\mathbf{e}$. By (28.2), that sum is $(\mu_{\mathrm{mv}} - r_f)/\sigma^2_{\mathrm{min}}$, so its sign is the sign of $\mu_{\mathrm{mv}} - r_f$.

RESULT **The tangency portfolio and the capital market line**

Let Σ be positive definite, $\boldsymbol{\mu}$ not proportional to $\mathbf{1}$, and $r_f < \mu_{\text{mv}}$. The fully invested portfolio with the highest Sharpe ratio is

$$\begin{aligned} \mathbf{w}^{\tan} &= \frac{\Sigma^{-1}\mathbf{e}}{\mathbf{1}^\top \Sigma^{-1}\mathbf{e}}, \\ \mathbf{1}^\top \Sigma^{-1}\mathbf{e} &= \frac{\mu_{\text{mv}} - r_f}{\sigma^2_{\min}} > 0, \\ \mathrm{SR}^{*2} &= \mathbf{e}^\top \Sigma^{-1}\mathbf{e}, \end{aligned} \tag{28.6}$$

with mean and volatility

$$\begin{aligned} \mu_{\tan} - r_f &= \frac{\mathrm{SR}^{*2}\sigma^2_{\min}}{\mu_{\text{mv}} - r_f}, \\ \sigma_{\tan} &= \frac{\mathrm{SR}^{*}\sigma^2_{\min}}{\mu_{\text{mv}} - r_f}. \end{aligned}$$

Its Sharpe ratio decomposes, and the capital market line passes through it:

$$\begin{aligned} \mathrm{SR}^{*2} &= \mathrm{SR}^2_{\infty} + \frac{(\mu_{\text{mv}} - r_f)^2}{\sigma^2_{\min}}, \\ \mu_p &= r_f + \mathrm{SR}^{*}\sigma_p. \end{aligned} \tag{28.7}$$

The portfolio $\mathbf{w}^{\tan}$ is the book on the direction $\Sigma^{-1}\mathbf{e}$ with $L = 1$, and its excess mean $\mathrm{SR}^{*2}/(\mathbf{1}^\top \Sigma^{-1}\mathbf{e})$ is positive. So it lies on the capital market line and attains SR^{*}, and its mean and volatility follow by substitution. Problem 8 derives (28.7) by splitting $\mathbf{e}$.

Every efficient book is a multiple of $\Sigma^{-1}\mathbf{e}$, so the mix of risky funds does not depend on how much of the book they take. Tobin (1958, read in Cowles Foundation Discussion Paper No. 14, 1956, §3.6) found this separation with risk-free cash at zero yield, no borrowing, and non-negativity assumed but not binding. Equation (28.6) allows any $r_f < \mu_{\text{mv}}$ and borrowing at r_f, and on the first window it shorts oil, which his non-negativity excludes.

By (28.6), $\Sigma\mathbf{w}^{\tan} = \mathbf{e}/(\mathbf{1}^\top \Sigma^{-1}\mathbf{e})$. So by (27.5) every fund's beta to the tangency portfolio is proportional to its excess return:

$$\begin{aligned} \beta_{i,\tan} &= \frac{e_i}{\mu_{\tan} - r_f}, \quad \text{that is,} \\ \mu_i - r_f &= \beta_{i,\tan}\,(\mu_{\tan} - r_f). \end{aligned}$$

The second form is the CAPM's equation of LM23, $\mathbb{E}[R_i] - r_f = \beta_i(\mathbb{E}[R_{\text{mkt}}] - r_f)$, with $\mathbf{w}^{\tan}$ in the market's place. Conversely, a fully invested portfolio with $\mu_p \neq r_f$ whose betas satisfy it for every fund is the normalized $\Sigma^{-1}\mathbf{e}$. The identity holds on any data, so it is no evidence for the CAPM. Why the market should be the tangency portfolio is a question of equilibrium theory, outside this book.

EXAMPLE 4

The tangency portfolio of the first window

Compute $\mathbf{w}^{\tan}$ at $r_f = 1.5\%$, check (28.7) and two-fund separation, and repeat at 0% and 2%.

SOLUTION

Here $\mu_{\mathrm{mv}} = 4.335\% > r_f$, so by (28.6)

$$\mathbf{1}^\top \Sigma^{-1}\mathbf{e} = \frac{0.02835}{0.0025152} = 11.27 > 0.$$

The tangency weights are $(11.8, 62.6, 37.8, 2.6, -14.7)\%$ (rounded separately; they sum to 100.1), with gross exposure 1.29. The mean is 8.376% and the volatility 7.810%, so $\mathrm{SR}^* = (8.376 - 1.5)/7.810 = 0.880$. The best single fund, 00646, has 0.72 from unrounded inputs, while Exhibit 1's rounded 10.0% and 11.9% would give 0.71. And (28.7) reads $0.7750 = 0.4554 + 0.5653^2$.

The efficient weights of Example 3 equal $\mathbf{w}^{\mathrm{mv}} + 0.905(\mathbf{w}^{\tan} - \mathbf{w}^{\mathrm{mv}})$, where $0.905 = 3.6565/4.0405$ is the ratio of the two portfolios' mean gaps to μ_{mv}, in points. By (27.5) every fund's beta to $\mathbf{w}^{\tan}$ equals $e_i/(\mu_{\tan} - r_f)$, to below 10^{-15}.

At $r_f = 0$ the weights are $(11.4, 48.7, 39.1, 10.3, -9.5)\%$, with $\mathrm{SR}^* = 1.10$. At 2% they are $(12.0, 71.1, 37.0, -2.2, -18.0)\%$ (rounded separately; they sum to 99.9), with 0.82. The illustrative rate moves 00646's weight by 22 points.

A risk-averse investor ranks books by their mean excess return minus a penalty on variance.

RESULT **Risk aversion with a risk-free asset**

Let $\delta > 0$ and Σ be positive definite, with borrowing and lending at r_f without limit. Then

$$\max_{\mathbf{f}} \ \mathbf{f}^\top \mathbf{e} - \frac{\delta}{2}\mathbf{f}^\top \Sigma \mathbf{f} \qquad \text{has the unique solution} \quad \mathbf{f}^* = \frac{\Sigma^{-1}\mathbf{e}}{\delta}, \tag{28.8}$$

with volatility SR^*/δ. When $r_f < \mu_{\mathrm{mv}}$, the solution is $\mathbf{f}^* = L\,\mathbf{w}^{\tan}$ with $L = \mathbf{1}^\top\Sigma^{-1}\mathbf{e}/\delta > 0$.

The objective's Hessian is $-\delta\Sigma$, so the objective is strictly concave for any r_f (LM10), and its one critical point is the maximum. By (10.8) that point solves $\mathbf{e} - \delta\Sigma\mathbf{f} = \mathbf{0}$, and its variance is $\mathbf{e}^\top\Sigma^{-1}\mathbf{e}/\delta^2$. The investor therefore holds the tangency mix, levered when $L > 1$ and diluted with cash when $L < 1$, and δ chooses only how much. LM7 found the same structure for two assets: its Kelly fractions are $L^*\mathbf{w}^{\tan}$. The Kelly fractions of N assets and growth-optimal leverage are LM43's.

EXAMPLE 5

A 10% volatility on the capital market line

On the first window, an investor targets 10% volatility a year, a hypothetical target. Find the mean, δ, L and the exposures.

SOLUTION

On the capital market line the mean is $1.5\% + 0.880 \times 10\% = 10.30\%$. By (28.8) the volatility SR^*/δ is 10% at $\delta = 0.880/0.10 = 8.80$, so $L = 11.27/8.80 = 1.28$, and the investor borrows 0.28 of capital. The exposures $\Sigma^{-1}\mathbf{e}/\delta$ are $(15.1, 80.1, 48.4, 3.3, -18.9)\%$ from unrounded inputs. The last is −18.860, while $1.28 \times (-14.7)$ would print −18.8. An investor with $\delta = 11.27$ holds the tangency portfolio and no cash.

PITFALL **No tangency portfolio when $r_f \geq \mu_{\mathrm{mv}}$**

In 22 of the through case's 92 windows, every month-end from 30 June 2022 to 30 April 2024 except 29 March 2024, μ_{mv} is below r_f. On 29 March 2024 itself $\mathbf{1}^\top\Sigma^{-1}\mathbf{e} = 0.614$ and the gross exposure is 29.1: as LM7 warned, a denominator near zero makes the weights explode.

Where $r_f > \mu_{\mathrm{mv}}$ the formula returns the worst portfolio, not the best. On 30 April 2024, $\mu_{\mathrm{mv}} = 1.34\%$ and $\mathbf{1}^\top\Sigma^{-1}\mathbf{e} = -0.185$, and the normalized $\Sigma^{-1}\mathbf{e}$ has Sharpe ratio −1.20 with gross exposure 86.0. The upper branch's Sharpe ratio rises toward $\mathrm{SR}_\infty = 1.2023$ without reaching it: 0.7864, 1.1928 and 1.2020 at 10, 100 and 1,000 points above μ_{mv}.

The through case's maximum-Sharpe rule R4 survives because it is long-only (LM29). A script that normalizes $\Sigma^{-1}\mathbf{e}$ without the sign test is wrong.

導讀 **切點投資組合與資本市場線**

加入無風險資產（risk-free asset）之後，所有效率投資組合（efficient portfolio）的曝險（exposure）都和 $\Sigma^{-1}\mathbf{e}$ 同方向，只差規模大小。把這個方向縮放成權重總和為 1，就是切點投資組合（tangency portfolio），它的夏普比率（Sharpe ratio）最高。從無風險利率（risk-free rate）出發、穿過切點的直線，就是資本市場線（capital market line）。前提是最小變異數投資組合（minimum-variance portfolio）的平均報酬（mean return）高於 r_f，低於 r_f 時同一個公式算出的是夏普比率最低的投資組合，等於 r_f 時則無法縮放（分母為零）。風險規避（risk aversion）係數 δ 只決定持有多少切點投資組合，δ 越大，現金越多。每檔基金對切點投資組合的 beta 都與它的超額報酬（excess return）成正比，這在任何資料上都成立，所以不能拿來驗證 CAPM。

KNOWLEDGE CHECK 3

In 22 of the 92 windows, r_f exceeds μ_{mv}. What does $\Sigma^{-1}\mathbf{e}/(\mathbf{1}^\top\Sigma^{-1}\mathbf{e})$ give there?

ANSWER

The denominator is negative, so the fully invested portfolio lies on the direction $-\Sigma^{-1}\mathbf{e}$. Its Sharpe ratio is $-\text{SR}^*$, the lowest of any fully invested portfolio: −1.20 on 30 April 2024, with gross exposure 86.0.

5 HOW TANGENCY WEIGHTS REACT TO THE MEANS

- ☐ compute how tangency weights respond to one mean, and compare that change with the mean's standard error
- ☐ explain why unconstrained mean–variance weights are in-sample estimates, not recommendations

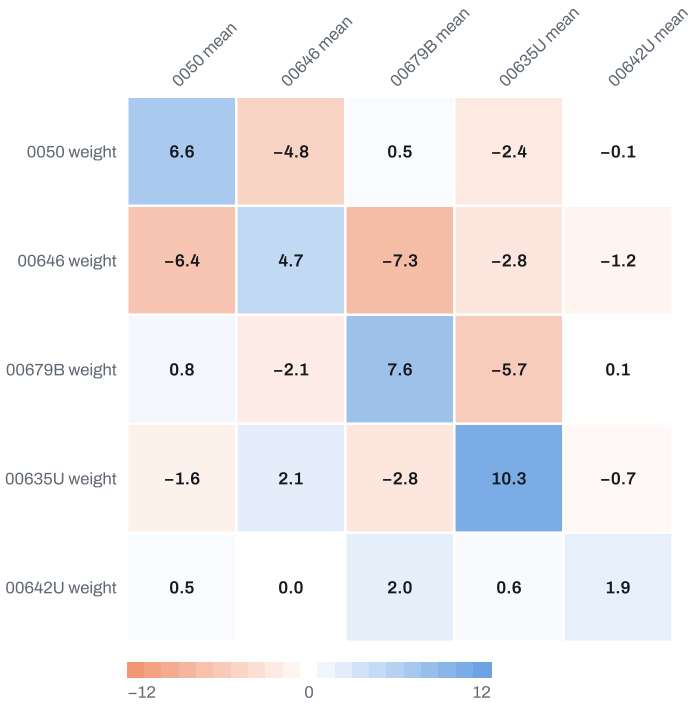
The means in (28.6) are estimates. Exposures respond to them linearly: by (28.8), a change $\Delta\mathbf{e}$ moves the exposures by exactly $\Sigma^{-1}\Delta\mathbf{e}/\delta$. By (11.8), Σ^{-1} divides each eigenvector's part of $\Delta\mathbf{e}$ by its eigenvalue, so a change along a direction of small variance moves the book the most.

The weights also divide by $\mathbf{1}^\top\Sigma^{-1}\mathbf{e}$, which moves too. A Jacobian matrix stacks as rows the gradients of several functions, each as in (10.8). Differentiating (28.6) by the quotient rule gives the weights' Jacobian, with entry (i, j) equal to $\partial w_i^{\tan}/\partial e_j$:

$$\begin{aligned} \Delta\mathbf{f}^* &= \frac{\Sigma^{-1}\Delta\mathbf{e}}{\delta} \text{ exactly,} \\ \frac{\partial\mathbf{w}^{\tan}}{\partial\mathbf{e}^\top} &= \frac{(I - \mathbf{w}^{\tan}\mathbf{1}^\top)\,\Sigma^{-1}}{\mathbf{1}^\top\Sigma^{-1}\mathbf{e}}. \end{aligned} \tag{28.9}$$

The Jacobian gives only a first-order approximation of a weight change. Exhibit 4 shows the exact changes on the first window: its diagonal runs from +1.9 points (oil) through +4.7 (00646), +6.6 (0050) and +7.6 (00679B) to +10.3 (gold).

Exhibit 4: Change in the tangency weights for a 1-point rise in one mean



Note: in-sample, first window, $r_f = 1.5\%$; percentage points. The cells are exact recomputations of (28.6), not (28.9)'s linear prediction, which gives 11.2 for gold's own weight. Each mean's standard error by (20.2), from unrounded inputs, is 9.4, 8.2, 5.9, 6.5 and 16.1 points, in the order of the columns. The cells are rounded separately, so 0050's and 00646's columns sum to −0.1, not 0. Source: as Exhibit 1. Computations: `data/lm28_charts.py`.

EXAMPLE 6

One more point on gold

Raise gold's mean by 1 point, from 0.78% to 1.78%, a hypothetical change of 0.15 of its standard error, 6.5 points. How do the tangency weights change, and the exposures at $\delta = 8.80$?

SOLUTION

Recomputing (28.6), gold's weight rises from 2.576% to 12.845%, by 10.3 points; 00679B's falls by 5.7 points and 00646's by 2.8. The Jacobian of (28.9) predicts 11.2 points for gold's own weight.

The exposures change exactly by column 4 of Σ^{-1} times $0.01/\delta$: $(-2.0, +3.2, -3.6, +14.6, -0.9)$ points, which sum to an 11.3-point rise in L.

Two years of means cannot fix tangency weights to the nearest 10 points: a change of 0.15 standard errors in gold's mean moves its weight by 10.3. Shrinking the means or the covariance matrix is the repair of LM30.

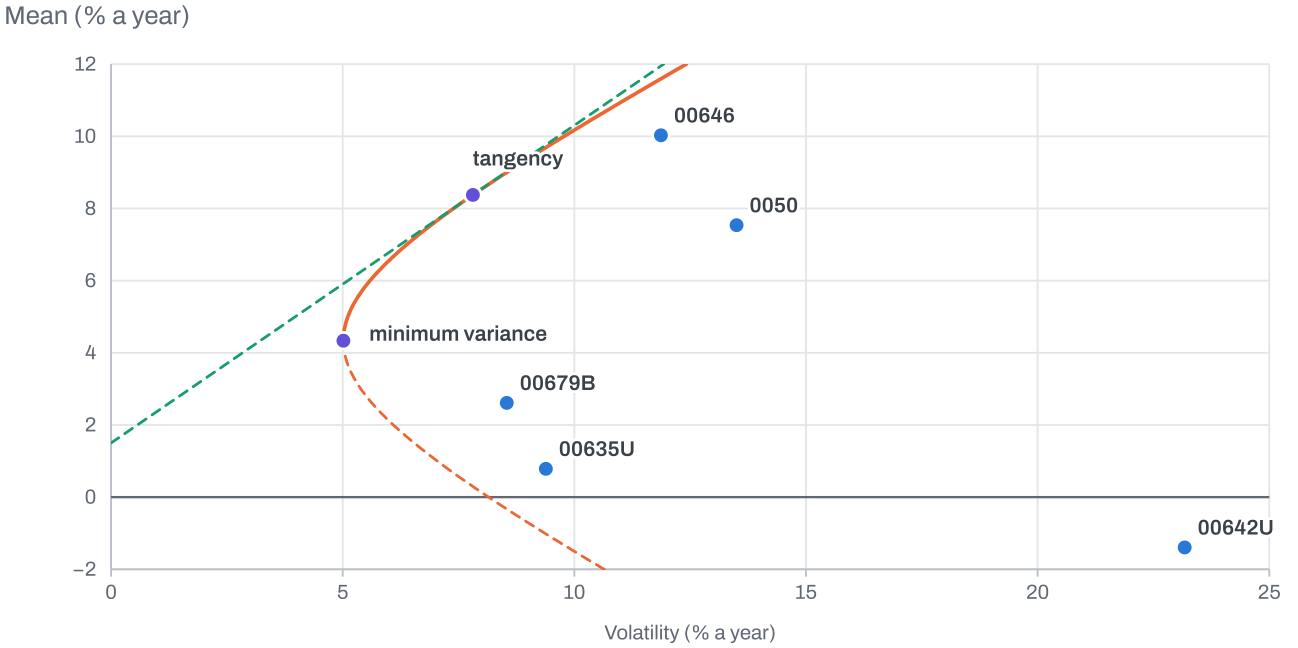
Best and Grauer (1991) find that with only a budget constraint an efficient portfolio's weights, mean and variance can be extremely sensitive to the means. LM30 measures how much more errors in means cost than errors in covariances.

The widget's first drawing uses the window's exact inputs and reproduces the funds, curves and two portfolios of Exhibit 3. Its sliders set 00646's mean, gold's mean and the correlation of 0050 and 00646. The window's Σ stays positive definite only for that correlation strictly between -0.736 and 0.969, so its slider reaches both failures.

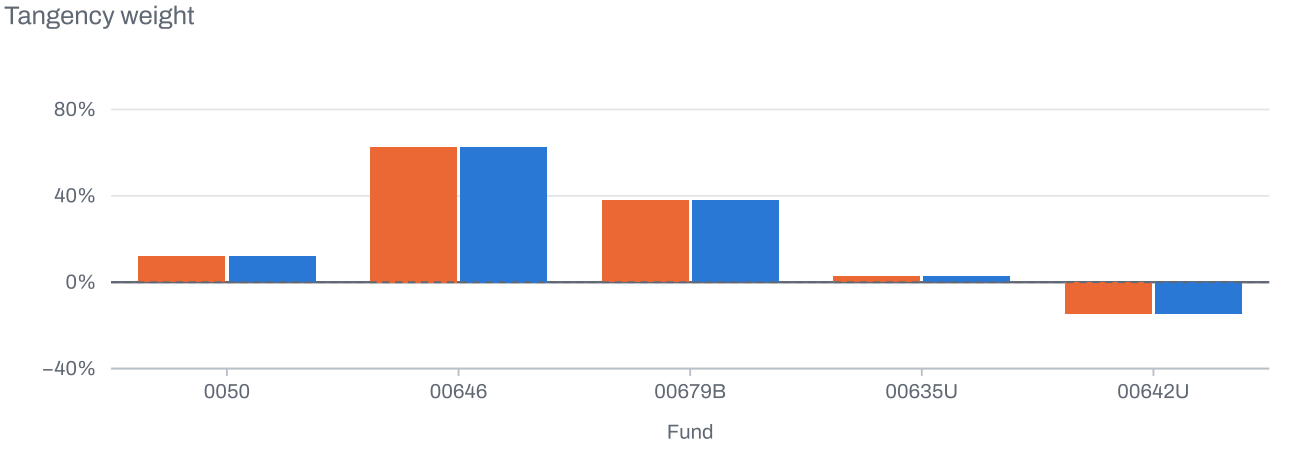
Each slider starts at a rounded value (10.0%, 0.8% and 0.54) and replaces its entry only once moved. With all three returned to their starting values, the readouts give $\mathbf{w}^{\tan} = (11.8, 62.4, 37.8, 2.7, -14.7)\%$, $\mathrm{SR}^{*} = 0.878$ and $\mu_{\mathrm{mv}} = 4.33\%$, against 62.6, 2.6, 0.880 and 4.34% for the window.

Frontier, tangency and one correlation

INTERACTIVE



first window with the sliders



w^{mv}	σ_{min}, a year	μ_{mv}, a year
(10.7, 22.5, 41.4, 25.0, 0.4)%	**5.02%**	**4.34%**
w^{tan}	SR*	Gross exposure of w^{tan}
(11.8, 62.6, 37.8, 2.6, −14.7)%	**0.880**	**1.29**

In-sample: the first window's plug-in moments, a risk-free rate of 1.5% a year. Solid: the efficient frontier, the minimum-variance frontier's upper half. Dashed: its lower half and, from 1.5%, the capital market line. The first drawing uses the window's exact inputs. The sliders move in rounded steps: once moved, a slider's value replaces the window's exact entry, so returning it to its starting value leaves the readouts slightly off the exhibit's values. Bars: tangency weights, the window's against the sliders'.

PITFALL **A derivative is local**

The Jacobian (28.9) overshoots gold's own change, 11.2 points against 10.3, because the sum $\mathbf{1}^\top \Sigma^{-1}\mathbf{e}$ grows with gold's mean and damps the change. Report exact recomputations, as Exhibit 4 does.

導讀 **平均數差一點，權重差很多**

曝險（exposure）$\Sigma^{-1}\mathbf{e}/\delta$ 對平均報酬（mean return）是線性的，但切點投資組合（tangency portfolio）的權重還要除以 $\mathbf{1}^\top \Sigma^{-1}\mathbf{e}$，所以是非線性的。敏感度（sensitivity）由 Σ^{-1} 決定：沿著變異數（variance）小的方向，平均數（mean）的一點變動會被放大很多。第一個視窗（window）中，黃金的平均報酬只要提高 1 個百分點，也就是標準誤（standard error）的 0.15 倍，它的權重就從 2.6% 升到 12.8%。兩年資料估出的平均數，估計誤差（estimation error）遠大於這個變動，所以樣本內（in-sample）算出的切點權重不能當成建議。LM30 討論修正的方法。

KNOWLEDGE CHECK 4

The exposures change linearly with $\mathbf{e}$, but the tangency weights do not. Why?

ANSWER

The weights divide the exposures by their total, $\mathbf{1}^\top \Sigma^{-1}\mathbf{e}/\delta$, which moves with $\mathbf{e}$ too. At $\delta = 8.80$ a 1-point rise in gold's mean raises L by 11.3 points, so every weight is divided by a larger total.

6 THE FIRST WINDOW HELD TO THE END

- ☐ split the gap between predicted and realized volatility into a moved minimum and a weight error
- ☐ explain why unconstrained mean–variance weights are in-sample estimates, not recommendations

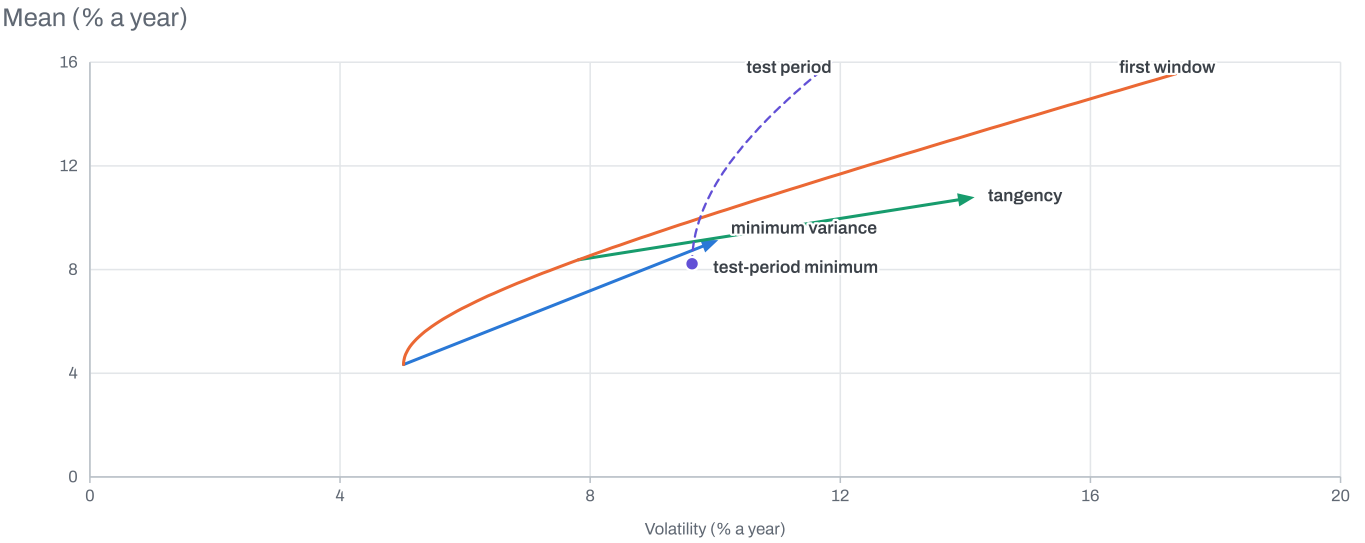
The test was fixed before any result. It holds $\mathbf{w}^{\text{mv}}$ and $\mathbf{w}^{\text{tan}}$ at constant weights from the close of 4 March 2019 to 8 October 2026, rebalanced every session, without costs.

The test period holds 1,851 returns of the wealth frame over 7.60 calendar years. The two portfolios' lowest values of $1 + \mathbf{w}^\top \mathbf{R}_t$ are 0.949 and 0.929, so the log growth is defined. On that frame 0050 earns 0

during its halt from 11 to 17 June 2025 and its six-session return on 18 June. Joint statistics use the 1,845 estimation rows, which lack those six sessions.

These two portfolios are not the through case's R3 and R4, their long-only forms re-solved every month with costs (LM29, LM34). Exhibit 5 draws the result.

Exhibit 5: Predicted against realized: the first window's portfolios held to the end



Note: an out-of-sample test of two portfolios fixed in advance, without costs. Arrows run from predicted to realized (volatility, mean), % a year, the realized points on the wealth frame. Solid: the first window's efficient frontier. Dashed: the test period's own frontier on its 1,845 joint dates (in-sample on later data), whose minimum has volatility 9.63% and mean 8.22%. The realized means' standard errors are 3.6 and 5.1 points: the vertical moves are within noise, the horizontal ones are not. Source: as Exhibit 1. Computations: `data/lm28_charts.py`.

EXAMPLE 7

Splitting the miss with (28.3)

On the 1,845 joint test dates, split each held portfolio's variance into the test period's minimum and a weight error.

SOLUTION

On the test rows, (28.2) gives $\sigma^2_{\min} = 0.00927$, so $\sigma_{\min} = 9.63\%$ against the window's 5.02%. By (28.3) and (10.4), the sample variance of a fixed mix splits exactly:

$$\begin{aligned} \mathbf{w}^{\mathrm{mv}} &: \; 0.00927 + 0.00081 = 0.01008, \\ \mathbf{w}^{\tan} &: \; 0.00927 + 0.01071 = 0.01998. \end{aligned}$$

The volatilities are 10.04% and 14.14%, so the weight errors add 0.41 and 4.51 points. Five decimals make the parts add as printed.

The wealth frame's 10.05% and 14.15% differ in the second decimal because its six more sessions raise the annualizer to $A_{\mathrm{w}} = 243.57$. On a common A, those sessions lower both volatilities slightly. The ex-post optimum moved too: $\mathbf{w}^{\mathrm{mv}} = (8.0, 22.4, 46.1, 17.0, 6.5)\%$, and $\mathbf{w}^{\tan} = (69.7, 8.8, -11.9, 27.6, 5.9)\%$ (rounded separately; it sums to 100.1).

EXAMPLE 8

Means and volatilities with their standard errors

Compare each held portfolio's realized volatility and mean with its prediction, in units of their standard errors.

SOLUTION

For $\mathbf{w}^{\mathrm{mv}}$ the realized volatility (divisor T) is 10.05%, with an iid standard error of 0.40 by LM20's $\sigma\sqrt{(\mathrm{Kurt}-1)/(4T)}$, the raw kurtosis 12.6 and $T = 1{,}851$. Its mean is 9.14% with standard error 3.6 by (20.2), and its log growth 8.63% a year. For $\mathbf{w}^{\tan}$ the three figures are 14.15% (standard error 0.53, from its unrounded kurtosis), 10.79% (standard error 5.1) and 9.79% a year.

The volatilities moved by 13 and 12 of their own standard errors, $(10.05 - 5.02)/0.40$ and $(14.15 - 7.81)/0.53$; the means by only 1.3 and 0.5. No realized Sharpe ratio is reported: how precisely it is known, and whether two differ, is LM33's question.

So the calculus and the weights were right in sample; the inputs changed, most of all the volatilities. Every fund's volatility rose (Exhibit 1), and the bond fund's negative correlations shrank on the joint dates: 0050–00679B from −0.30 to −0.09 and 00646–00679B from −0.26 to −0.06. With the window's correlations, the test volatilities alone would raise the minimum from 5.02% to 8.35%, most of the way to 9.63%.

Large squared returns cluster, so the iid standard error of a volatility is probably too small (LM20). For normal days the two standard errors would be 0.17 and 0.23. Gaps of 13 and 12 standard errors survive any reasonable inflation of the error. The first window was the third calmest of 92, so the miss is partly the luck of the split. A second split would be a second test.

PITFALL **A frontier fitted to the past is not an opportunity set**

The window's frontier described two calm years, and the test period's own frontier lies far to its right (Exhibit 5). A predicted volatility at an in-sample optimum is also biased low, because the optimizer favours directions where $\hat{\Sigma}$ is small. LM30 measures that bias. It is not the main cause here, since the minimum itself moved from 5.02% to 9.63%.

導讀 **樣本內最適，樣本外落空**

用第一個視窗（window）估出的兩個投資組合，事先固定權重，一路持有到 2026 年 10 月，這是樣本外（out-of-sample）檢驗，視窗內的計算則是樣本內（in-sample）。最小變異數投資組合（minimum-variance portfolio）實現的波動率（volatility）是 10.05%，是預測的兩倍，但測試期在共同交易日上的最小值就有 9.63%，權重估錯只多了 0.41 個百分點。換句話說，計算與權重在樣本內都正確，是輸入值變了：每一檔基金的波動都上升，債券基金的負相關係數（correlation coefficient）也變弱。平均報酬（mean return）的偏離只有 1.3 和 0.5 個標準誤（standard error），仍在雜訊範圍內；波動率的偏離卻有 12 到 13 個標準誤。

KNOWLEDGE CHECK 5

Which part of Example 7's miss could any choice of weights have avoided?

ANSWER

Only the weight error, 0.41 points for $\mathbf{w}^{\mathrm{mv}}$ and 4.51 for $\mathbf{w}^{\tan}$. By (28.3) the moved minimum, from 5.02% to 9.63%, binds every fully invested portfolio on the 1,845 joint test dates.

7 MEAN–VARIANCE IN PYTHON

- [ ] compute frontier portfolios in NumPy from linear systems, and check them and their multipliers with cvxpy

The KKT system of (28.1) stacks stationarity, $2\Sigma\mathbf{w} = K^{\top}\boldsymbol{\lambda}$, and feasibility into one linear system:

$$\begin{pmatrix} 2\Sigma & K^{\top} \\ K & 0 \end{pmatrix} \begin{pmatrix} \mathbf{w} \\ -\boldsymbol{\lambda} \end{pmatrix} = \begin{pmatrix} \mathbf{0} \\ \mathbf{b} \end{pmatrix}.$$

PYTHON

```python
import cvxpy as cp
import numpy as np

# X: the first window's 504 x 5 daily returns; A = 242.95, rf = 0.015;
# mu_p: a target mean. Keep mu, S and rf in the same (annual) units:
# w_tan depends on them, w_mv does not.
mu = A * X.mean(axis=0)
S = A * np.cov(X, rowvar=False, ddof=0)
np.linalg.cholesky(S)                    # raises LinAlgError if not PD
one = np.ones(5)
x = np.linalg.solve(S, one)
w_mv = x / x.sum()                       # (28.2)
y = np.linalg.solve(S, mu - rf * one)
assert y.sum() > 0                       # r_f < mu_mv, (28.6)
w_tan = y / y.sum()

K = np.vstack([one, mu])                 # a frontier point, (28.1)
b = np.array([1.0, mu_p])
KKT = np.block([[2 * S, K.T], [K, np.zeros((2, 2))]])
w, neg_lam = np.split(np.linalg.solve(KKT, np.r_[np.zeros(5), b]), [5])

v = cp.Variable(5)
cons = [one @ v == 1, mu @ v == mu_p]
prob = cp.Problem(cp.Minimize(cp.quad_form(v, cp.psd_wrap(S))), cons)
prob.solve(solver=cp.CLARABEL, tol_gap_abs=1e-12, tol_gap_rel=1e-12,
           tol_feas=1e-12)
print(prob.status, np.abs(v.value - w).max(), cons[0].dual_value)
```

EXAMPLE 9

A frontier point by the KKT system and by cvxpy

At the efficient mean of Example 3, 7.99% (unrounded), solve the KKT system and check it with cvxpy.

SOLUTION

The KKT system reproduces the closed forms (28.1) and (28.4) to below 10^{-15}. cvxpy 1.9.3 then solves the problem on the annualized inputs, with a fresh `Problem` per solver. OSQP 1.1.3 runs with `eps_abs` and `eps_rel` 10^{-10} and polishing, and CLARABEL 0.11.1 with `tol_gap_abs`, `tol_gap_rel` and `tol_feas` 10^{-12}.

Both return status `optimal`. Their weights differ from the closed form by below 10^{-15} (OSQP) and below 10^{-11} (CLARABEL, an interior-point method). The KKT residuals, stationarity with cvxpy's duals and feasibility, are below 10^{-13}.

The two equality duals are +0.00193 and −0.161, the negatives of Example 3's λ and λ_μ, which follow LM7's signs. That sign is the installed package's behaviour. The cvxpy documentation gives no sign convention for equalities, so check the sign against a closed form.

LM29 adds inequality constraints, which need a solver. Boyd and Vandenberghe (2004) give Markowitz portfolio optimization (§4.4.1) and its risk–return trade-off (§4.7.6) as optimization examples.

IN PYTHON **Companion lab**

The Polars-first notebook `LM28_lab.ipynb` rebuilds the module's exhibits and examples. It computes the minimum-variance and tangency portfolios by hand, the frontier, the sensitivity to the means, the held portfolios and their split, and the KKT system with cvxpy. It scans the 92 windows against the through case's R3, solves Problems 10 to 12, and ends by adding the funds one at a time.

The notebook reads FinMind files that you download once with your own token. From the repository root, run `uv run python data/fetch_finmind.py`, then `uv run python data/make_lab_data.py`. FinMind's licence does not allow the book to redistribute the data. After that, the notebook runs offline.

KNOWLEDGE CHECK 6

Why `np.linalg.solve(S, ones)` rather than `np.linalg.inv(S) @ ones`? And what does a failed `np.linalg.cholesky(S)` tell you before any solve?

ANSWER

A solve factorizes Σ once and rounds less than forming the inverse and multiplying. A failed Cholesky factor means that Σ is not positive definite (LM12): some nonzero weight vector has zero or negative variance, and (28.1) need not have a unique minimum.

SUMMARY

- For a positive definite Σ and linear constraints, minimum variance has one closed-form solution, (28.1), whose identity proves uniqueness.
- The minimum-variance portfolio (28.2) has covariance $\sigma^2_{\min}$ with every fund. On E5's first window it predicts 5.02%.
- The frontier (28.4)–(28.5) is a parabola in variance and mean, and any two frontier portfolios rebuild it (two-fund separation).
- Beside a risk-free asset every efficient book is a multiple of $\Sigma^{-1}\mathbf{e}$. The tangency portfolio (28.6) exists only when $r_f < \mu_{\text{mv}}$, and δ in (28.8) chooses how much of it to hold.
- Tangency weights are fragile: 0.15 standard errors on gold's mean move its weight by 10.3 points.
- Held to 2026, the minimum-variance portfolio missed mostly because the minimum itself moved, from 5.02% to 9.63%. The tangency portfolio's weight error added 4.51 points more.
- Unconstrained mean–variance weights are in-sample estimates, not recommendations. LM29 adds constraints, and LM30 repairs the inputs.

PRACTICE PROBLEMS

Problems on data use E5's first window: 504 estimation-frame returns from 13 February 2017 to 27 February 2019, with $A = 242.95$ and divisor T. The inputs are annual, $\boldsymbol{\mu} = A\bar{\mathbf{R}}$ and $\Sigma = A\hat{\Sigma}$, and $r_f = 1.5\%$ a year, simple. Portfolios are fully invested and otherwise unconstrained unless a stem says otherwise, and weights are in percent to one decimal.

1. Three uncorrelated assets have volatilities 10%, 20% and 20% (constructed). The first asset's weight in the minimum-variance portfolio is *closest* to:

   A. 50.0%

   B. 66.7%

   C. 33.3%

2. Two assets have volatilities 20% and 10% and correlation $\rho = 0.6$ (constructed). The first asset's weight in the minimum-variance portfolio is *closest* to:

   A. −7.7%

   B. 0%

   C. 20.0%

3. A frontier has $\sigma_{\min} = 5\%$, $\mu_{\text{mv}} = 4\%$ and $\text{SR}_\infty = 0.5$ (constructed). The volatility of the frontier portfolio with mean 8% is *closest* to:

   A. 13.0%

   B. 8.0%

   C. 9.43%

4. For the frontier of Problem 3 and $r_f = 1.5\%$, the best Sharpe ratio SR^* is *closest* to:

   A. 0.50

   B. 0.707

   C. 1.00

5. Beside a risk-free asset, $\mathrm{SR}^* = 0.8$, $\mathbf{1}^\top \Sigma^{-1} \mathbf{e} = 8$ in annual units and $\delta = 4$ (constructed). The book's volatility and its total exposure L are *closest* to:

   A. 20% and 2.0

   B. 20% and 0.5

   C. 320% and 2.0

6. A frontier has $\sigma_{\min} = 5\%$, $\mu_{\mathrm{mv}} = 0\%$ and $\mathrm{SR}_\infty = 0.5$, and $r_f = 1.5\%$ (constructed). The Sharpe ratio of the portfolio $\Sigma^{-1}\mathbf{e}/(\mathbf{1}^\top \Sigma^{-1}\mathbf{e})$ is *closest* to:

   A. +0.58

   B. −0.50

   C. −0.58

7. Derive (28.1) from the Lagrangian and (10.8). Prove its identity with (10.9) and deduce uniqueness. Then specialize to (28.2) and $\lambda^* = 2\sigma^2_{\min}$.

8. (a) Derive (28.5) from (28.4). (b) Show that the frontier approaches the asymptotes $\mu_p = \mu_{\mathrm{mv}} \pm \mathrm{SR}_\infty \sigma_p$. (c) Derive (28.7) by splitting $\mathbf{e} = (\boldsymbol{\mu} - \mu_{\mathrm{mv}}\mathbf{1}) + (\mu_{\mathrm{mv}} - r_f)\mathbf{1}$ and showing that the cross term is zero. (d) For $r_f < \mu_{\mathrm{mv}}$, write every frontier portfolio as a point of the line through $\mathbf{w}^{\mathrm{mv}}$ and $\mathbf{w}^{\tan}$.

9. With $N = 2$, show that (28.2) is LM7's (7.7) and (7.8). With LM7's 2017–2021 inputs, $\sigma_1^2 = 0.026800$, $\sigma_2^2 = 0.018353$ and $\sigma_{12} = -0.004908$, compute the first weight, $\sigma_{\min}$ and λ^*.

10. On the first window: (a) Raise 0050's mean by 1 point and recompute $\mathbf{w}^{\tan}$, giving each change in points. (b) Explain the two largest changes with the window's 0.54 correlation of 0050 and 00646 and with column 1 of Σ^{-1}. (c) Compare 1 point with 0050's standard error, and say what the data do and do not support.

11. *(Python)* On the first window, compute (28.2), (28.6), and (28.4) at the efficient mean with equal weights' volatility, from unrounded inputs, with `np.linalg.solve`. Solve the KKT system at that mean. Check both with cvxpy on the annualized inputs, a fresh `Problem` per solver: OSQP with `eps_abs` and `eps_rel` 10^{-10} and polishing, and CLARABEL with `tol_gap_abs`, `tol_gap_rel` and `tol_feas` 10^{-12}. Report the statuses, the largest weight gap, the KKT residuals and the duals' signs.

12. *(Python, out-of-sample)* Hold the first window's $\mathbf{w}^{\text{mv}}$ at constant weights, rebalanced every session without costs, on the wealth frame's 1,851 returns from 5 March 2019 to 8 October 2026. Compute its realized volatility (divisor T) with $A_{\text{w}} = 243.57$, and its iid standard error by LM20's formula with the raw kurtosis. Then split its variance by (28.3) on the estimation frame's 1,845 test rows, with divisor T and $A = 242.95$.

SOLUTIONS

1. **B** is correct: with uncorrelated assets $\Sigma^{-1}\mathbf{1} = (100, 25, 25)$, so the weights are inverse variances, $100/150 = 66.7\%$, and $\sigma_{\min} = \sqrt{1/150} = 8.16\%$. A weights by inverse volatilities, and C ignores the volatilities.

2. **A** is correct: by (7.7) the weight is $(0.01 - 0.012)/(0.04 + 0.01 - 0.024) = -7.7\%$. It shorts the riskier asset because $\rho > \sigma_2/\sigma_1 = 0.5$ (LM7). B assumes a long-only answer, and C uses inverse variances, ignoring the correlation.

3. **C** is correct: by (28.5), $\sigma_p^2 = 0.05^2 + 0.04^2/0.5^2 = 0.0089$, so $\sigma_p = 9.43\%$. A adds volatilities, $5\% + 0.04/0.5$, and B drops $\sigma_{\min}$.

4. **B** is correct: by (28.7), $\text{SR}^{*2} = 0.5^2 + (0.025/0.05)^2 = 0.5$, so $\text{SR}^* = 0.707$, and by (28.6) $\mu_{\tan} = 1.5\% + 0.5 \times 0.0025/0.025 = 6.50\%$. A is the asymptote's slope, and C adds the two terms unsquared.

5. **A** is correct: by (28.8) the volatility is $\text{SR}^*/\delta = 20\%$ and $L = 8/4 = 2.0$. B inverts the ratio, $\delta/8$, and C multiplies SR^* by δ.

6. **C** is correct: the split of Problem 8(c) holds for any r_f, so $\text{SR}^{*2} = 0.5^2 + 0.015^2/0.05^2 = 0.34$ and $\text{SR}^* = 0.583$. The denominator is negative, so the portfolio lies on $-\Sigma^{-1}\mathbf{e}$ with Sharpe ratio $-\text{SR}^*$, the lowest possible. On the window of 30 April 2024 the same formula gave −1.20. A ignores the sign of the denominator, and B is minus the asymptote's slope.

7. Section 2 gives the derivation: the gradient (10.8) vanishes at $\mathbf{w} = \tfrac{1}{2}\Sigma^{-1}K^\top\boldsymbol{\lambda}$, the constraint fixes $\boldsymbol{\lambda}$, and (10.9) gives the identity. With $K = \mathbf{1}^\top$ and $\mathbf{b} = 1$, the multiplier is $2/(\mathbf{1}^\top\Sigma^{-1}\mathbf{1}) = 2\sigma_{\min}^2$ and $\mathbf{w}^* = \sigma_{\min}^2\Sigma^{-1}\mathbf{1}$, which is (28.2).

8. (a) In (28.4), $\boldsymbol{\Delta}$ is $(\mu_p - \mu_{\text{mv}})/\text{SR}_\infty^2$ times $\Sigma^{-1}(\boldsymbol{\mu} - \mu_{\text{mv}}\mathbf{1})$, and $\mathbf{1}^\top\boldsymbol{\Delta} = 0$ (Section 3). That direction's quadratic form is SR_∞^2, so by (28.3) the variance is $\sigma_{\min}^2 + (\mu_p - \mu_{\text{mv}})^2/\text{SR}_\infty^2$, which is (28.5). (b) By (28.5), $\mu_p - \mu_{\text{mv}} = \pm\text{SR}_\infty\sqrt{\sigma_p^2 - \sigma_{\min}^2}$, whose distance from $\pm\text{SR}_\infty\sigma_p$ is $\text{SR}_\infty\sigma_{\min}^2/(\sigma_p + \sqrt{\sigma_p^2 - \sigma_{\min}^2})$. That distance tends to 0 as σ_p grows. (c) The split gives

$$\begin{aligned}\mathbf{e}^\top\Sigma^{-1}\mathbf{e} = \text{SR}_\infty^2 \\ + 2(\mu_{\text{mv}} - r_f)\,\mathbf{1}^\top\Sigma^{-1}(\boldsymbol{\mu} - \mu_{\text{mv}}\mathbf{1}) \\ + (\mu_{\text{mv}} - r_f)^2\,\mathbf{1}^\top\Sigma^{-1}\mathbf{1}.\end{aligned}$$

   The cross term is zero by Section 3, and $\mathbf{1}^\top\Sigma^{-1}\mathbf{1} = 1/\sigma_{\min}^2$. (d) The same split, divided by $\mathbf{1}^\top\Sigma^{-1}\mathbf{e} = (\mu_{\text{mv}} - r_f)/\sigma_{\min}^2$, gives

$$\mathbf{w}^{\tan} = \mathbf{w}^{\mathrm{mv}} + \frac{\sigma_{\min}^2}{\mu_{\mathrm{mv}} - r_f}\,\Sigma^{-1}(\boldsymbol{\mu} - \mu_{\mathrm{mv}}\mathbf{1}),$$

the frontier portfolio (28.4) at $\mu_{\tan}$. So (28.4) reads

$$\mathbf{w}(\mu_p) = \mathbf{w}^{\mathrm{mv}} + \frac{\mu_p - \mu_{\mathrm{mv}}}{\mu_{\tan} - \mu_{\mathrm{mv}}}\,(\mathbf{w}^{\tan} - \mathbf{w}^{\mathrm{mv}}),$$

with the coefficient 0.905 in Example 4.

9. By (9.5), $\Sigma^{-1}\mathbf{1} = (\sigma_2^2 - \sigma_{12},\ \sigma_1^2 - \sigma_{12})^{\top}/\det\Sigma$, whose entries sum to $(\sigma_1^2 + \sigma_2^2 - 2\sigma_{12})/\det\Sigma$. So the first weight is (7.7), and $\sigma_{\min}^2 = \det\Sigma/(\sigma_1^2 + \sigma_2^2 - 2\sigma_{12})$ is (7.8). With LM7's inputs, $0.023261/0.054969 = 42.3\%$, $\sigma_{\min} = 9.22\%$ and $\lambda^* = 0.0170$, as LM7 printed.

10. (a) The changes are +6.6 points in 0050, −6.4 in 00646, and +0.8, −1.6 and +0.5 in the others (Exhibit 4's first column). (b) By (28.9) the exposures move along column 1 of Σ^{-1}, $(82.8, -48.1, 25.8, -17.3, -0.7)$ in annual units: up in 0050 and down in 00646, its closest substitute at correlation 0.54. The weights then divide by a larger total: unrounded, column 1 sums to 42.6, so $\mathbf{1}^{\top}\Sigma^{-1}\mathbf{e}$ rises from 11.27 to 11.70. Column 1 over the new total moves 0050 by +7.08 points and 00646 by −4.11. Rescaling the old weights takes 0.43 and 2.28 more, leaving 6.6498 and −6.3892 unrounded. (c) One point is only 0.11 of 0050's standard error, 9.4 points, yet it moves two weights by more than 6 points. The data support the window's minimum-variance mix far better than any tangency weight, which they cannot fix to within 10 points.

11. The lab reproduces Example 1, Example 4, Example 3 and Example 9. Both solvers return `optimal`, with weight gaps to the closed form below 10^{-15} (OSQP) and 10^{-11} (CLARABEL) and KKT residuals below 10^{-13}. The duals +0.00193 and −0.161 are the negatives of λ and λ_μ in Example 3.

12. The lab gives 10.05% on the wealth frame, with standard error 0.40. On the 1,845 test rows the variance splits as $0.00927 + 0.00081 = 0.01008$, that is $\sigma_{\min} = 9.63\%$ plus 0.41 points of weight error, 10.04% in all. The wealth frame's six more sessions raise the annualizer to A_{w}, hence its 10.05% (Example 8, Example 7).

GLOSSARY

Term	中文	Meaning
Minimum-variance portfolio	最小變異數投資組合	The fully invested portfolio of least variance, $\mathbf{w}^{\mathrm{mv}}$, (28.2).
Minimum-variance frontier	最小變異數前緣	The fully invested portfolios of least variance for each target mean, (28.4)–(28.5).
Efficient frontier	效率前緣	The upper half of the minimum-variance frontier.
Efficient portfolio	效率投資組合	A portfolio on the efficient frontier, or, beside a risk-free asset, on the capital market line.
Two-fund separation	兩基金分離	Any two frontier portfolios rebuild the whole frontier, because its weights are affine in the target mean.
Tangency portfolio	切點投資組合	The fully invested portfolio with the highest Sharpe ratio, $\Sigma^{-1}\mathbf{e}/(\mathbf{1}^{\top}\Sigma^{-1}\mathbf{e})$ when $r_f < \mu_{\mathrm{mv}}$, (28.6).
Capital market line	資本市場線	The line $\mu_p = r_f + \mathrm{SR}^{*}\sigma_p$ of efficient books beside a risk-free asset, (28.7).
Risk-free asset	無風險資產	An asset with the certain return r_f, here an illustrative 1.5% a year.
Risk aversion	風險規避	The weight δ of the variance penalty in (28.8).
Shadow price	影子價格	A multiplier read as the rate at which the optimal value changes with its constraint, (7.6).
Sensitivity	敏感度	The change of an optimal weight for a change of an input, exact or by the Jacobian (28.9).
Jacobian matrix	Jacobi 矩陣	The matrix whose rows are the gradients of several functions.
In-sample / out-of-sample	樣本內／樣本外	Data used to estimate a portfolio versus later data used to test it.

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